NARASARAOPETA ENGINEERING COLLEGE (AUTONOMOUS)

Official Study Material • Database Management Systems (DBMS) (CS301) • Unit 3: Normalization & Database Design

Course Code:	CS301	**Regulation:**	R20 / R23 Autonomous
Subject:	Database Management Systems (DBMS)	**Unit:**	Unit 3 - Normalization & Database Design

NARASARAOPETA ENGINEERING COLLEGE (AUTONOMOUS)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

ACADEMIC STUDY MATERIAL: CS301 DATABASE MANAGEMENT SYSTEMS

UNIT 3: NORMALIZATION & DATABASE DESIGN

Regulation: R20 / R23 Autonomous Syllabus | Credits: 4

1. PURPOSE OF NORMALIZATION & ANOMALIES:

Normalization organizes database tables to minimize data redundancy and prevent update, insertion, and deletion anomalies.

- Insertion Anomaly: Inability to insert information without adding unrelated attributes (e.g. cannot add course without enrolled student).

- Deletion Anomaly: Unintended loss of crucial information upon deleting a record (e.g. deleting last student deletes course information).

- Update Anomaly: Inconsistent data when a duplicated value is changed in one row but not others.

2. FUNCTIONAL DEPENDENCY (FD) & CLOSURE ALGORITHM:

An FD X -> Y holds on relation R if whenever two tuples agree on attribute X, they must also agree on attribute Y.

- Armstrong's Axioms:

1. Reflexivity: If Y $\subseteq$ X, then X -> Y.

2. Augmentation: If X -> Y, then XZ -> YZ.

3. Transitivity: If X -> Y and Y -> Z, then X -> Z.

- Attribute Closure Algorithm (X+):

Calculates all attributes functionally determined by attribute set X.

Step 1: Set X+ = X.

Step 2: For each FD A -> B in F, if A $\subseteq$ X+, add B to X+.

Step 3: Repeat until X+ does not change.

If X+ contains all attributes of R, then X is a Super Key of R.

3. NORMAL FORMS (1NF to BCNF):

- First Normal Form (1NF): Every attribute domain contains only atomic (indivisible) values. No multi-valued or composite attributes.

- Second Normal Form (2NF): Must be in 1NF and have NO Partial Dependency (no non-prime attribute depends on a proper subset of any candidate key).

- Third Normal Form (3NF): Must be in 2NF and have NO Transitive Dependency. For every non-trivial FD X -> Y, either:

* X is a Super Key, OR

* Y is a Prime Attribute (member of some candidate key).

- Boyce-Codd Normal Form (BCNF): Strict 3NF form. For every non-trivial FD X -> Y:

* X MUST BE A SUPER KEY.

Every BCNF relation is in 3NF, but not all 3NF relations are in BCNF.

4. DECOMPOSITION PROPERTIES & TRADE-OFFS:

- Lossless-Join Decomposition: A decomposition of R into R1 and R2 is lossless if and only if R1 ∩ R2 -> R1 or R1 ∩ R2 -> R2 (common attributes form a superkey of at least one relation).

- Dependency Preservation: A decomposition preserves dependencies if the union of FDs of decomposed relations covers the original FD set.

- 3NF vs BCNF Trade-off: 3NF always guarantees both Lossless Join and Dependency Preservation. BCNF guarantees Lossless Join and eliminates all redundancy based on FDs, but cannot always preserve functional dependencies.

5. IMPORTANT EXAM QUESTIONS & ANSWERS:

Q1: Differentiate between 3NF and BCNF with a suitable example.

Answer: In 3NF, for X -> Y, Y can be a prime attribute even if X is not a superkey. In BCNF, X must strictly be a superkey. Example: Relation R(Student, Subject, Teacher) with FDs { (Student, Subject) -> Teacher, Teacher -> Subject }. Candidate keys are (Student, Subject) and (Student, Teacher). Teacher -> Subject violates BCNF because Teacher is not a superkey, but satisfies 3NF because Subject is a prime attribute.

Q2: Given R(A, B, C, D, E) and F = { A -> BC, CD -> E, B -> D, E -> A }. Find candidate keys and highest normal form.

Answer: Closure of A: A+ = {A, B, C, D, E}. A is candidate key. E+ = {E, A, B, C, D}. E is candidate key. BC+ = {B, C, D, E, A}. BC is candidate key. CD+ = {C, D, E, A, B}. CD is candidate key. B -> D violates 2NF because D is non-prime and depends on B (subset of key BC). Therefore, highest normal form is 1NF.